Literature Review: Deep Learning for Seizure Detection and Prediction from Non-EEG Wearable Signals (2015–2025)(Exposé)

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Contents

1 Introduction

1.1 Problem Statement and Motivation

Artificial intelligence (AI) and deep learning have made significant improvements regarding fast and reliable decision making on a large set of information. This ability is esppecialy helpful when trying to detect specific patterns The focus of this literature review is on time-series deep learning models (1D-CNN/LSTM/TCN) applied to wearable seizure detection/prediction using non-EEG wearable/autonomic biosignals: Electrocardiography

(ECG), Heart Rate Variability (HRV), Photoplethysmography (PPG), Electrodermal Activity (EDA), and Accelerometry (ACC). Extended temporal context, novel deep learning architectures, training strategies, and personalized fine-tuning can improve performance metrics (sensitivity, AUC, f1-score, false alarms per hour (FAR/h), latency) in clinical and ambulatory contexts.

Epilepsy affects roughly 7.6 per 1,000 people and about 30% of patients are drug-resistant, creating demand for timely, low-burden monitoring and early intervention (Beghi, 2019; Chen et al., 2020; Kwan & Brodie, 2000). While EEG remains the reference in controlled settings, it is bulky, uncomfortable, and requires complex setup, and therefore not suitable for ambulatory monitoring. Wearables offer the possibility of continuous monitoring in ambulatory settings (Chen et al., 2020).

Healthcare constraints directly affect modeling objectives. Models must achieve high sensitivity while controlling false alarms per hour (FAR/h) to avoid alarm fatigue. They must also balance predictive performance with interpretability and safety requirements demanded in regulated clinical contexts. Robustness to domain shift across patients, devices, sites, and activities is required, and evaluations should address data scarcity by prioritizing prospective, multimodal, patient-preserving study designs.

This review covers time-series deep learning approaches to a focused medical use case, comparing traditional feature-based models and emerging deep temporal architectures (1D-CNN, LSTM, TCN, attention/transformers) for seizure detection and preictal prediction from non-EEG wearable biosignals.

The key question is which modeling and evaluation choices deliver clinically viable, low FAR/h, high-sensitivity performance under pseudo-prospective and prospective, ambulatory conditions.

Cross-dataset transfer is essential to assess robustness and domain shift (Andrade et al., 2024; Yang et al., 2022).

1.2 Research Aim and Questions

This literature review focuses on machine/deep learning models for seizure detection and prediction using non-EEG wearable/autonomic biosignals, and aims to identify the optimal strategies to achieve clinically meaningful performance in ambulatory contexts.

1.2.1 Primary Research Question

In non-EEG wearable seizure detection and prediction (ECG/HRV, PPG, EDA, ACC), which modeling and evaluation choices reported in 2015–2025 are associated with low alarm-based false-alarm rates and high sensitivity with acceptable latency in ambulatory settings?

1.2.2 Sub-questions

1. What ranges of sensitivity, FAR/h, and latency are reported under retrospective, pseudo-prospective, and prospective designs, and how do alarm-based results differ from sample-based metrics?

2. Across studies, how do traditional feature-based models compare to 1D-CNN/TCN/LSTM, and what evidence exists for attention/ transformer variants on these signals, including any reported computational/latency considerations?

3. How do temporal choices (window length, stride, context length) correlate with reported alarm-based performance in real-time use?

4. What evidence is available on robustness to domain shift (patient/device/site/activity), including cross-dataset/external validation and how do these affect alarm-based metrics?

5. What dataset gaps and class imbalance issues are reported, and what is the measured impact of augmentation or synthetic data on alarm-based outcomes?

Contribution. The goal is to synthesize model designs for wearable seizure detection/prediction, analyze how evaluation choices affect reported performance and derive practical recommendations to achieve low FAR/h while keeping high sensitivity in ambulatory deployment.

1.3 Methodological Approach and Structure

The review follows structured IS/medical literature guidance (concept matrix inspired by (Webster & Watson, 2002) and PRISMA principles). The methodological approach uses a multi-stage search and transparent extraction/synthesis procedures as outlined below.

1. **Search strategy (Multi-stage, 2015–2025):**
   - *Scoping:* Google Scholar for broad coverage and identifying candidate papers.
   - *Formal queries:* IEEE Xplore, PubMed, and Scopus using controlled keyword strings and deduplication.
   - *Citation chasing:* Backward and forward citation tracking on key papers to ensure completeness.

2. **Inclusion / Exclusion criteria:**
   - *Inclusion:* Studies using wearable/non-EEG autonomic signals for seizure detection or preictal prediction.
   - *Exclusion:* EEG-only studies, invasive recordings, or papers limited to purely algorithmic simulations without human data.

Extracted dimensions populate a concept matrix with the following fields:

1. Setting: clinical vs ambulatory.

2. Modality: ECG/HRV, PPG, EDA, ACC.

3. Task: detection vs prediction.

4. Model class and fusion strategy.

5. Study phase: retrospective, pseudo-prospective, prospective.

6. Evaluation granularity: sample- vs alarm-based.

7. Windowing specifics (window length, stride, overlap).

8. Metrics: sensitivity, FAR/h, AUC, f1-score, latency.
9. Personalization.
10. Interpretability and safety considerations.
11. Data characteristics and dataset identity.
12. Missing data handling (imputation, exclusion) and synthetic data use.
13. Inference requirements (latency, computational load, hardware).

Prioritization Criteria: Studies including splits preserving patient-specific data, FAR/h reporting and external or cross validation will be prioritized in the synthesis. A PRISMA-style flow diagram will summarize and visualize screening and inclusion.

1.4 Provisional Table of Contents

Target Length: Approx. 15 Pages of Text

1. **Introduction** (approx. 1.5 pages)
   - 1.1 Problem Statement and Motivation
   - 1.2 Research Aim and Questions
   - 1.3 Structure of this Review
2. **Methodological Approach** (approx. 1.5 pages)
   - 2.1 Search Strategy (Multi-stage, 2015–2025)
   - 2.2 Inclusion and Exclusion Criteria
   - 2.3 Extraction Dimensions and Concept Matrix
   - 2.4 PRISMA Flow Diagram
3. **Theoretical Background and Clinical Concepts** (approx. 2 pages)
   - 3.1 Clinical Deployment Requirements (Sensitivity vs. FAR/h, Alarm Fatigue)
   - 3.2 Seizure Classification and Clinical Implications
   - 3.3 Autonomic Markers and Wearable Sensing (ECG/HRV, PPG, EDA, ACC)
   - 3.4 Study Phases: Retrospective vs. Pseudo-Prospective vs. Prospective
4. **Modeling Landscape and Design Dimensions** (approx. 5 pages)
   - 4.1 Architectural Evolution: Feature Engineering vs. Deep Learning
     - Traditional Pipelines (SVM, RF, kNN with HRV/EDA Features)
     - Deep Temporal Models (1D-CNN, LSTM, TCN)
     - Attention and Transformer Variants
   - 4.2 Sensor Modalities and Fusion Strategies
     - Feature-Level, Decision-Level, and Representation-Level Fusion
     - Complementary Cues under Noise and Non-Stationarity

- 4.3 Personalization and Domain Shift
  - Patient-Specific Fine-Tuning and Calibration
  - Cross-Dataset Evaluation and External Validation
- 4.4 Robustness and Generalization
  - Artifact Handling and Augmentation Strategies
  - Out-of-Distribution Detection
- 4.5 Deployment Constraints (Latency, Energy, Interpretability)

5. **Results: Synthesis and Comparative Analysis** (approx. 3 pages)
   - 5.1 Performance Ranges by Study Phase and Setting
   - 5.2 Model Comparison: Feature-Based vs. Deep Learning
   - 5.3 Impact of Temporal Choices (Window Length, Stride, Context)
   - 5.4 Evidence on Robustness and Domain Shift
   - 5.5 Dataset Gaps, Class Imbalance, and Augmentation Effects
6. **Discussion** (approx. 1.5 pages)
   - 6.1 Trade-offs: Complexity vs. Latency vs. Interpretability
   - 6.2 Clinical vs. Ambulatory Deployment Considerations
   - 6.3 Limitations of Current Evidence
7. **Conclusion and Future Work** (approx. 0.5 pages)
   - 7.1 Summary of Key Findings
   - 7.2 Practical Recommendations
   - 7.3 Research Gaps and Future Directions

Note: The following sections provide preliminary background to contextualize the research questions. They will be expanded with systematic literature findings in the final paper.

2 Theoretical Background and Clinical Concepts

2.1 Clinical Deployment Requirements

In epilepsy monitoring, sensitivity (detecting all seizures to ensure safety) must be weighed against precision (minimizing false alarms). In a clinical ward, high sensitivity is prioritized as medical staff can verify alarms. In ambulatory (home) settings, however, a high FAR/h causes "alarm fatigue," rendering the device useless for the patient.

Finally, the "Black Box" nature of deep learning poses safety challenges. Unlike decision trees or logistic regression, deep neural networks often lack transparency. For clinical acceptance, systems generally require explainability to support physician judgment and meet regulatory requirements.

2.2 Seizure Classification and Clinical Implications

Epileptic seizures are classified by their onset as focal (starting in a specific area of the brain) or generalized (affecting both hemispheres from the onset).

Focal seizures are further subdivided into focal aware and focal impaired awareness seizures, depending on whether the patient is aware during the seizure or not, and by onset features (motor vs non-motor). Focal seizures may evolve to bilateral tonic–clonic seizures.

Generalized seizures begin in both cerebral hemispheres and include motor (e.g. tonic–clonic, myoclonic, atonic) and non-motor (absence) types (Fisher et al., 2017). Tonic–clonic seizures commonly cause loss of consciousness and carry increased injury and SUDEP (Sudden Unexpected Death in Epilepsy) risk (Hesdorffer et al., 2011).

2.3 Autonomic Markers and Wearable Sensing

Sympathetic activation around seizures manifests as ictal tachycardia, preictal HR rise with reduced HRV (lower RMSSD/HF, higher LF/HF), increased EDA (tonic SCL, phasic SCRs), and limited ACC changes useful for artifacts/context. PPG yields pulse-rate variability analogous to HRV from ECG (Mason et al., 2024; Miron et al., 2025). Wearable heart-rate sensing aligns closely with chest-strap ECG under motion (e.g., Apple Watch Ultra/Polar H10 concordance ∼0.998), though dominant-wrist motion degrades accuracy. Chest-strap ECG provides robust RR intervals during exercise (Wolf et al., 2024).

2.4 Study Phases: Retrospective vs. Pseudo-Prospective vs. Prospective

Retrospective studies use pre-recorded data, giving the model access to the full dataset including future samples. Pseudo-prospective designs simulate real-time performance by processing data sequentially without future knowledge, providing a more realistic performance estimate. Prospective studies collect and process data in true real-time, representing the highest standard for clinical validity and the scenario in which alarm-based metrics (FAR/h) matter most.

3 Modeling Landscape and Design Dimensions

Detection and preictal prediction from ECG/HRV, PPG, EDA, and ACC are framed by windowing, labeling, and alarm logic, with alarm-based sensitivity, FAR/h, and latency as the outcomes used in the research questions.

3.1 Architectural Evolution: Feature Engineering vs. Deep Learning

Classical pipelines utilizing HRV, EDA, and ACC features combined with SVM, RF, or kNN remain competitive in retrospective settings (Mason et al., 2024; Seth et al., 2023). However, deep temporal models such as 1D-CNN, LSTM, and TCN often improve phase discrimination. Recently, attention variants and Transformers are being explored on these signals (Abtahi et al., 2025; Kalousios et al., 2024; Pordoy et al., 2025; Yang et al., 2022). A key inquiry of this review is determining how feature-based models compare to deep models and whether attention mechanisms add distinct value under ambulatory, alarm-based evaluation.

3.2 Sensor Modalities and Fusion Strategies

Fusion across ECG, EDA, ACC, and PPG at the feature, decision, or representation level seeks complementary cues and is relevant to maintaining performance under noise and non-stationarity (Pordoy et al., 2025; Yang et al., 2022).

3.3 Personalization and Domain Shift

Patient-specific fine-tuning and post-hoc calibration shift the operating point toward lower FAR/h and higher sensitivity in ambulatory use (Andrade et al., 2024). Thresholding and periodic recalibration are commonly used to handle drift.

3.4 Robustness and Generalization

Robustness requires explicit checks of domain shift across patient, device, site, and activity. Cross-dataset evaluation (e.g., training on TUH and testing on RPAH or EPILEPSIAE) exposes this shift and favors alarm-based over sample-based metrics (Andrade et al., 2024; Yang et al., 2022). Artifact handling, augmentation through time-warping, jitter, or scaling, and Out-of-Distribution (OOD) detection support generalization (Kalousios et al., 2024; Seth et al., 2023).

3.5 Deployment Constraints: Latency, Energy, and Interpretability

Wearable seizure detection imposes strict deployment constraints. Detection latency must remain below clinically relevant thresholds (typically seconds to allow caregiver response or patient warning), which limits model complexity and inference time. Energy consumption determines battery life; continuous inference on resource-constrained MCUs or mobile processors requires lightweight architectures such as quantized CNNs or efficient RNNs. Compact sequence models (e.g., depthwise-separable 1D-CNNs, temporal convolutional networks) and event-driven or on-demand processing reduce computational burden.

Interpretability remains critical for clinical adoption. Regulators and clinicians require transparency in alarm generation to trust automated decisions affecting patient safety. Post-hoc explanation methods (e.g., attention weights, Grad-CAM on temporal segments, SHAP values for HRV features) can highlight which signal segments or features triggered an alarm. Hybrid pipelines that combine interpretable HRV/EDA features with deep representations offer a middle ground between performance and explainability.

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